DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture
(6^{th} SEMESTER, 4^{th} Year) LAB EXPERIMENT # 03

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Score: ____________ Signature of the Lab Tutor: ______________ Date: ____________

Getting Familiar with LACP

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives:

As a result of this lab section, you should achieve the following tasks:

- Manually set the line rate on an interface
- Configuration of manual mode link aggregation
- Configuration of link aggregation using static LACP mode
- Management of the priority of interfaces in static LACP mode

Keywords: Link Aggregation, Eth-Trunk	**Duration:** 03 hours

1 Overview of lab

1.1. Introduction

This lab focuses on the configuration and verification of Link Aggregation Control Protocol (LACP) using Huawei's eNSP (Enterprise Network Simulation Platform). LACP, defined under IEEE 802.3ad, is used to bundle multiple physical links between network devices into a single logical link to achieve redundancy and increased bandwidth.

The purpose of this lab is to demonstrate how LACP works in a simulated Huawei network environment by configuring link aggregation between switches, verifying operational status, and observing the failover behavior when a link fails. This enhances understanding of network resiliency, load balancing, and high availability concepts.

1.2. LACP:

It is a protocol defined in the IEEE 802.3ad standard that allows you to combine multiple physical network links into a single logical link for increased bandwidth and redundancy.

Aggregates multiple Ethernet interfaces into one logical link to:

- Increase bandwidth
- Provide failovers (if one link fails, traffic continues) Easily add/remove links without downtime.
- LACP automates link negotiation and maintenance.

1.3. LACP Modes Active Mode:

In this mode, the device actively sends LACP packets to negotiate and form a link aggregation. Ensures that aggregation can be formed even if the other side is passive

Command: mode lacp-static

Passive Mode:

The device waits for LACP packets from the other side. It responds but does not initiate. Works only if the other side is in active mode.

LACP will only form a link aggregation if at least one side is in Active mode if both sides are passive no aggregation will occur.

1.4. Eth-Trunk

An Eth-Trunk (Ethernet Trunk) in Huawei devices is a logical interface that combines multiple physical Ethernet links into a single aggregated link.

- You first create an Eth-Trunk interface
- Then you bind multiple physical interfaces to that trunk
- Use mode lacp-static to enable LACP negotiation
- Switch decides which ports can bundle based on LACP status

Command: [Huawei] interface Eth-Trunk1

Tools Required

- Huawei Switches **(**Virtual in eNSP)

– S5700 or any other L2/L3 switch model that supports LACP. LACP-supported Interfaces

– Ensure switches have available GE (Gigabit Ethernet) ports for aggregation.

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Topology

Scenario

As a network administrator of an existing enterprise network, it has been requested that the connections between the switches be used more effectively by preparing the switches to support link aggregation before establishing manual link aggregation, for which the media between the switches are to be configured as member links.

Tasks

Step 1 Perform basic configuration on Ethernet switches

Auto-negotiation is enabled on Huawei switch interfaces by default. The rate of G0/0/1 and G0/0/2 on S1 and S2 are to be set manually.

Change the system name and view detailed information for G0/0/1 and G0/0/2 on S1.

```
<Huawei> undo terminal monitor
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S1
[S1]
[S1] display interface GigabitEthernet 0/0/1
GigabitEthernet0/0/1 current state : UP
Line protocol current state : UP Description:
Switch Port, PVID :    1, TPID : 8100(Hex), The Maximum Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-cc88-13e7
Last physical up time   : 2020-05-08 21:51:40 UTC-08:00
Last physical down time : 2020-05-08 21:51:39 UTC-08:00
Current system time: 2020-05-08 21:52:23-08:00
Hardware address is 4c1f-cc88-13e7
    Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
    Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
    Input: 238 bytes, 2 packets
    Output: 2618 bytes, 22 packets
    Input:
      Unicast: 0 packets, Multicast: 2 packets
      Broadcast: 0 packets
Output:
```

```
      Unicast: 0 packets, Multicast: 22 packets
      Broadcast: 0 packets
    Input bandwidth utilization  :    0%
Output bandwidth utilization :    0%

[S1] display interface GigabitEthernet 0/0/2
GigabitEthernet0/0/2 current state : UP
Line protocol current state : UP Description:
Switch Port, PVID :    1, TPID : 8100(Hex), The Maximum Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-cc88-13e7
Last physical up time   : 2020-05-08 21:51:40 UTC-08:00
Last physical down time : 2020-05-08 21:51:39 UTC-08:00
Current system time: 2020-05-08 21:52:38-08:00
Hardware address is 4c1f-cc88-13e7
    Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
    Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
    Input: 238 bytes, 2 packets
Output: 3451 bytes, 29 packets
Input:
      Unicast: 0 packets, Multicast: 2 packets
      Broadcast: 0 packets
Output:
      Unicast: 0 packets, Multicast: 29 packets
      Broadcast: 0 packets
    Input bandwidth utilization  :    0%
Output bandwidth utilization :    0%

[S1]
```

Set the rate of G0/0/1 and G0/0/2 on S1 to 100 Mbit/s.
Before changing the interface rate, disable auto-negotiation

```
[S1]
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] undo negotiation auto
[S1-GigabitEthernet0/0/1] speed 100
[S1-GigabitEthernet0/0/1] q
[S1]
[S1] interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] undo negotiation auto
[S1-GigabitEthernet0/0/2] speed 100
[S1-GigabitEthernet0/0/2] q
[S1]
```

Set the rate of G0/0/1 and G0/0/2 on S2 to 100 Mbit/s.

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```
<Huawei>
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S2
[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] undo negotiation auto
[S2-GigabitEthernet0/0/1] speed 100
[S2-GigabitEthernet0/0/1] display this
[S2-GigabitEthernet0/0/1] q
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] undo negotiation auto
[S2-GigabitEthernet0/0/2] speed 100
[S2-GigabitEthernet0/0/2] display this
[S2-GigabitEthernet0/0/2] q
[S2]
```

Confirm that the rate of G0/0/1 and G0/0/2 have been set on S1.

```
[S1] display interface GigabitEthernet 0/0/1
GigabitEthernet0/0/1 current state : UP
Line protocol current state : UP Description:
Switch Port, PVID :    1, TPID : 8100(Hex), The Maximum Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-cc88-13e7
Last physical up time   : 2020-05-08 22:02:15 UTC-08:00
Last physical down time : 2020-05-08 21:56:06 UTC-08:00
Current system time: 2020-05-08 22:03:50-08:00
Hardware address is 4c1f-cc88-13e7
    Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
    Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
    Input: 357 bytes, 3 packets
Output: 39627 bytes, 333 packets
Input:
      Unicast: 0 packets, Multicast: 3 packets
      Broadcast: 0 packets
Output:
      Unicast: 0 packets, Multicast: 333 packets
      Broadcast: 0 packets
    Input bandwidth utilization  :    0%
    Output bandwidth utilization :    0%

[S1]
[S1] display interface GigabitEthernet 0/0/2
GigabitEthernet0/0/2 current state : UP
Line protocol current state : UP Description:
```

```
Switch Port, TPID : 8100(Hex), The Maximum Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-cc88-13e7
Last physical up time   : 2020-05-08 21:59:14 UTC-08:00
Last physical down time : 2020-05-08 21:51:39 UTC-08:00
Current system time: 2020-05-08 22:15:34-08:00
Hardware address is 4c1f-cc88-13e7
    Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
    Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
    Input: 4998 bytes, 42 packets
    Output: 0 bytes, 0 packets
Input:
      Unicast: 0 packets, Multicast: 42 packets
      Broadcast: 0 packets
Output:
      Unicast: 0 packets, Multicast: 0 packets
      Broadcast: 0 packets
    Input bandwidth utilization  :    0%
Output bandwidth utilization :    0%

[S1]
```

Step 2 Configure manual link aggregation

Create Eth-Trunk 1 on S1 and S2.

```
[S1]
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] q
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] eth-trunk 1
[S1-GigabitEthernet0/0/1] q
[S1]
[S1] interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] eth-trunk 1
[S1-GigabitEthernet0/0/2] q
[S1]

[S2]
[S2] interface Eth-Trunk 1
[S2-Eth-Trunk1] q
[S2]
[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] eth-trunk 1
[S2-GigabitEthernet0/0/1] q
[S2]
```

```
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] eth-trunk 1
[S2-GigabitEthernet0/0/2] q
[S2]
```

Verify the Eth-Trunk Configuration

```
[S2] display eth-trunk 1
Eth-Trunk1's state information is:
WorkingMode: NORMAL         Hash arithmetic: According to SIP-XOR-DIP
Least Active-linknumber: 1  Max Bandwidth-affected-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
PortName                      Status      Weight GigabitEthernet0/0/1
Up          1        GigabitEthernet0/0/2          Up          1

[S2]

[S1] display eth-trunk 1
Eth-Trunk1's state information is:
WorkingMode: NORMAL         Hash arithmetic: According to SIP-XOR-DIP
Least Active-linknumber: 1  Max Bandwidth-affected-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
PortName                      Status      Weight GigabitEthernet0/0/1
Up          1        GigabitEthernet0/0/2          Up          1

[S1]
```

The grey highlighted lines indicate that the Eth-Trunk works properly.

Step 3 Configure Link Aggregation in Static LACP Mode

Delete the configurations from G0/0/1 and G0/0/2 on S1 and S2.

```
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] undo eth-trunk
[S1-GigabitEthernet0/0/1] q
[S1]
[S1] interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] undo eth-trunk
[S1-GigabitEthernet0/0/2] q
[S1]

[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] undo eth-trunk
[S2-GigabitEthernet0/0/1] q
```

```
[S2]
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] undo eth-trunk
[S2-GigabitEthernet0/0/2] q
[S2]
```

Create Eth-Trunk 1 and set the load balancing mode of the Eth-Trunk to static LACP mode.

```
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] mode lacp-static
[S1-Eth-Trunk1] q
[S1]
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] eth-trunk 1
[S1-GigabitEthernet0/0/1] q
[S1]
[S1] interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] eth-trunk 1
[S1-GigabitEthernet0/0/2] q
[S1]

[S2]
[S2] interface Eth-Trunk 1
[S2-Eth-Trunk1] mode lacp-static
[S2-Eth-Trunk1] q
[S2]
[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] eth-trunk 1
[S2-GigabitEthernet0/0/1] q
[S2]
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] eth-trunk 1
[S2-GigabitEthernet0/0/2] q
[S2]
```

Verify that the LACP-static mode has been enabled on the two links.

```
[S1] display eth-trunk
Eth-Trunk1's state information is:
Local:
LAG  ID:  1                                      WorkingMode:  STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System   Priority:   32768                  System   ID:   4c1f-cc88-13e7
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
```

```
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight

GigabitEthernet0/0/1   Selected 1GE      32768   2      305     10111100  1
GigabitEthernet0/0/2   Selected 1GE      32768   3      305     10111100  1
Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/1   32768    4c1f-cccf-03af  32768   2      305     10111100
GigabitEthernet0/0/2   32768    4c1f-cccf-03af  32768   3      305     10111100

[S1]
```

Set the system priority on S1 to 100 to ensure S1 remains the Actor.

```
[S1]
[S1] lacp priority 100
[S1]
```

Set the priority of the interface and determine active links on S1.

```
[S1]
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] lacp priority 100
[S1-GigabitEthernet0/0/1] q
[S1] interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] lacp priority 100
[S1-GigabitEthernet0/0/2] q
[S1]
```

Verify the Eth-Trunk configuration.

```
[S1] display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                    WorkingMode: STATIC
Preempt Delay: Disabled      Hash arithmetic: According to SIP-XOR-DIP
System Priority: 100         System ID: 4c1f-cc88-13e7
Least Active-linknumber: 1   Max Active-linknumber: 8
Operate status: up           Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/1   Selected 1GE      100     2      305     10111100  1
GigabitEthernet0/0/2   Selected 1GE      100     3      305     10111100  1
Partner:
--------------------------------------------------------------------------------
```

```
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/1   32768    4c1f-cccf-03af  32768   2      305     10111100
GigabitEthernet0/0/2   32768    4c1f-cccf-03af  32768   3      305     10111100

[S1]

[S2] display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                   WorkingMode: STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768      System ID: 4c1f-cccf-03af
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/1   Selected 1GE      32768   2      305     10111100  1
GigabitEthernet0/0/2   Selected 1GE      32768   3      305     10111100  1
Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/1   100      4c1f-cc88-13e7  100     2      305     10111100
GigabitEthernet0/0/2   100      4c1f-cc88-13e7  100     3      305     10111100

[S2]
```

Final Configuration

```
[S1] display current-configuration
#
!Software Version V200R008C00SPC500
sysname S1 #
lacp priority 100
#
interface Eth-Trunk1
mode lacp #
interface GigabitEthernet0/0/1
eth-trunk 1 lacp priority 100
undo negotiation auto speed
100 #
interface GigabitEthernet0/0/2 eth-trunk
1
lacp priority 100
undo negotiation auto
speed 100 # return
```

```
[S2]display current-configuration
#
!Software Version V200R008C00SPC500
sysname S2 #
interface Eth-Trunk1
mode lacp #
interface GigabitEthernet0/0/1
eth-trunk 1 undo negotiation
auto speed 100 #
interface GigabitEthernet0/0/2
eth-trunk 1 undo negotiation
auto speed 100 # return
```

Step 4 Save the topology and configuration

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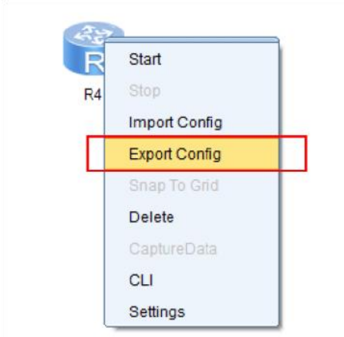
Upload the .topo .efz .cfg files run the **save** command to save the current configuration on each device

```
<Huawei> save
```

- **Export** the devices' configuration
- click the **Save/Save as** button on the Tools bar to save the whole project.



Tasks

Try the "`export config`" feature to generate `.cfg` file for each device. Also try "`import config`" feature too.

Implement Link Aggregation and try transferring file from one PC to another to demonstrate the difference in throughput and transfer time.

Upload the `.topo .efz .cfg` files

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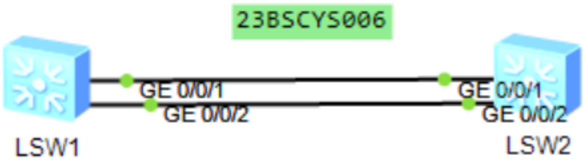


```
Eth-Trunkl's state information is:
Local:
LAG ID: 1                    WorkingMode: STATIC
Preempt Delay: Disabled      Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768       System ID: 4c1f-cc7d-0d7e
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up           Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/1   Selected 1GE      32768   2      305     10111100  1
GigabitEthernet0/0/2   Selected 1GE      32768   3      305     10111100  1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/1   100      4c1f-cc33-3e71  100     2      305     10111100
GigabitEthernet0/0/2   100      4c1f-cc33-3e71  100     3      305     10111100
```

```
sysname SW-1
#
lacp priority 100
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface MEth0/0/1
#
interface Eth-Trunk1
 mode lacp-static
#
interface GigabitEthernet0/0/1
 undo negotiation auto
 speed 100
 eth-trunk 1
 lacp priority 100
#
interface GigabitEthernet0/0/2
 undo negotiation auto
```